

RSFQ-JEPA: A Neural Surrogate for SPICE-Level Screening of Superconducting RSFQ Logic Cells

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Abstract

We present RSFQ-JEPA, a neural surrogate that predicts SPICE-level electrical properties and design-rule violations of rapid single-flux-quantum (RSFQ) logic cells directly from their layout and circuit-graph representation. Inference runs in milliseconds with calibrated per-prediction uncertainty, roughly 1,000x to 3,000x faster than SPICE per cell. RSFQ-JEPA is an honest v0, validated against JoSIM on the open ColdFlux SFQ5ee+ cell library. It is intended as a fast first-pass screen for chip-level verification and design-space search, with low-confidence predictions automatically flagged for SPICE fallback.

1 Introduction

RSFQ (single-flux-quantum) verification relies on SPICE-class simulation, typically JoSIM [1], to confirm functional correctness and to extract operating margins for each cell. A single cell margin simulation takes minutes. That cost does not scale to chip-level screening or to design-space search, where thousands of cell variants must be evaluated. A fast, uncertainty-aware surrogate makes both screening and gradient-based exploration tractable, provided it reports honestly when it is uncertain.

2 Approach

RSFQ-JEPA uses a hierarchical JEPA (joint-embedding predictive architecture) [3] backbone that combines a graph neural network over the cell’s circuit graph with a vision transformer over the cell’s layout raster, joined by cross-attention. The model has 297.5M parameters and 17 prediction heads covering propagation delay, peak current, functional correctness, design-rule violations, and operating margins. Each head produces a per-prediction uncertainty estimate via MC-dropout with split-conformal calibration, so that every output carries a calibrated bound.

The model is trained on the open ColdFlux SFQ5ee+ cell library from Stellenbosch University [2], using JoSIM-generated ground-truth labels and evaluated against the MIT-LL SFQ5ee+ design rules [4]. We describe the method at a high level here; the focus of this note is the validated accuracy and speed of the resulting surrogate.

3 Results

Table 1 reports accuracy on held-out cells against JoSIM ground truth. We tabulate the heads with direct JoSIM ground truth; the remaining heads are bounded scores or quantities learned from analytic oracles, which lack a single-number JoSIM RMSE and are not listed here.

RSFQ-JEPA is roughly 1,000x to 3,000x faster than a JoSIM cell simulation per cell: warm inference takes about 0.2 s, versus several minutes for a JoSIM margin characterization. Every

Table 1: Held-out accuracy of RSFQ-JEPA against JoSIM ground truth on the ColdFlux SFQ5ee+ cell library.

Quantity	Result
Propagation delay	14 ps RMSE
Peak current	2 uA RMSE
Functional classification	approximately 100%
Design-rule-violation detection	99.2%
Margin heads (Ic margin, bias margin, max frequency)	3 to 4% MAE
Conformal coverage ($\alpha = 0.1$)	at least 89%

prediction carries a calibrated 1-sigma bound, and low-confidence predictions auto-flag for SPICE fallback, so the surrogate is used as a screen rather than a replacement for the oracle.

4 Limitations

This is a v0. The reported accuracy is surrogate-versus-oracle (versus JoSIM), not fab-calibrated or silicon-validated. The model predicts cell properties; it does not design cells. Its scope is the SFQ5ee+ cell family.

5 Availability

Code, the cell schema, example cells, and benchmarks are at <https://github.com/CogitanAI/RSFQ-JEPA>. A live interactive demo is at <https://demo.cogitan.ai>, with more on the broader Fluxus suite at <https://cogitan.ai/fluxus>. Access and a technical brief are available on request (dylan@cogitan.ai). Model weights are not publicly released. RSFQ-JEPA is the deep-screening engine behind Cogitan’s Fluxus superconducting design suite.

References

- [1] J. Delport et al., JoSIM, a superconducting circuit simulator.
- [2] L. Schindler, T. Hall, and C. Fourie, ColdFlux RSFQlib, Stellenbosch University.
- [3] Y. LeCun, A Path Towards Autonomous Machine Intelligence: the Joint-Embedding Predictive Architecture (JEPA).
- [4] S. Tolpygo et al., MIT Lincoln Laboratory SFQ5ee+ fabrication process.